



US 20250283604A1

(19) **United States**

(12) **Patent Application Publication**
MILLER, III

(10) **Pub. No.: US 2025/0283604 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **INTERLOCKING ANDIRON SYSTEMS AND
RELATED METHODS**

(52) **U.S. Cl.**
CPC **F24B 1/193** (2013.01)

(71) Applicant: **Southern Andiron and Tool Co.,**
Atlanta, GA (US)

(72) Inventor: **William T. MILLER, III**, Brookhaven,
GA (US)

(21) Appl. No.: **18/595,804**

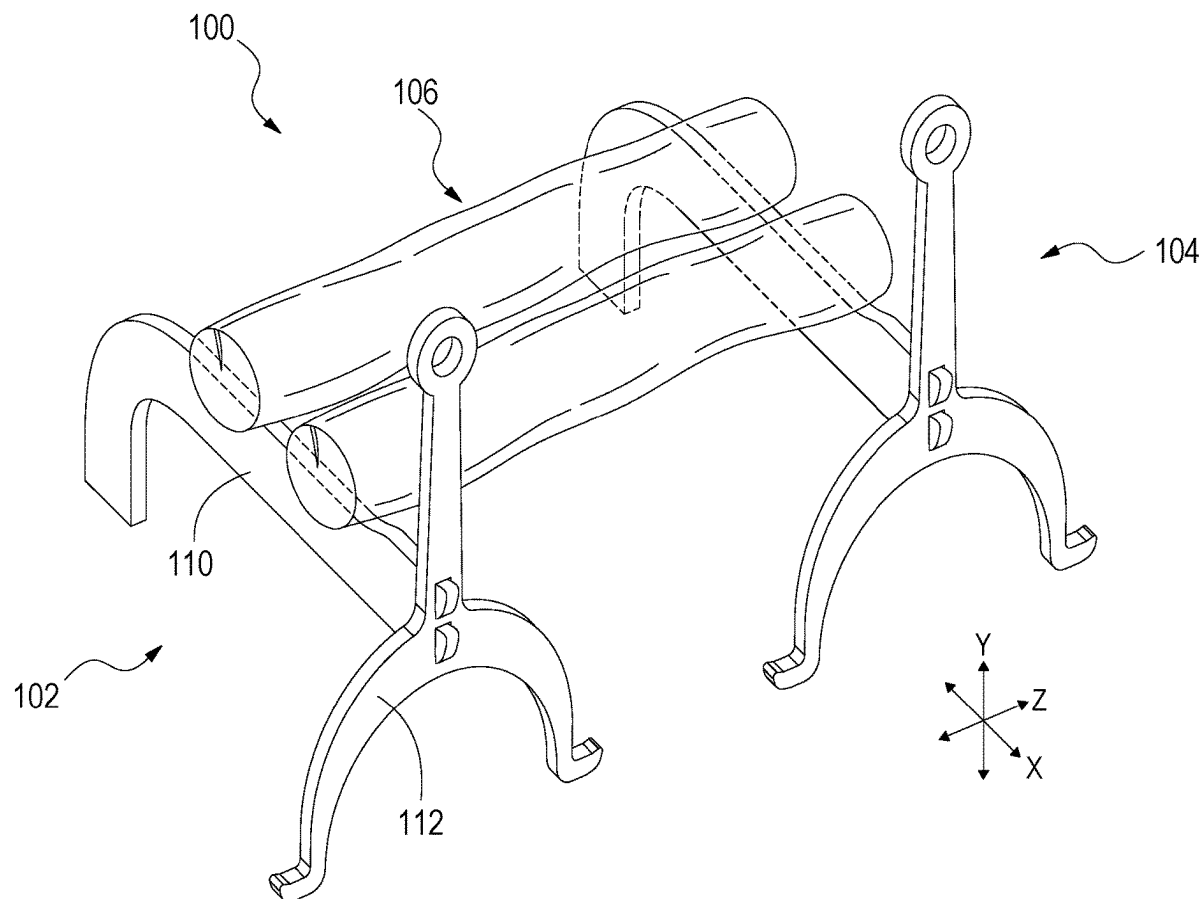
(22) Filed: **Mar. 5, 2024**

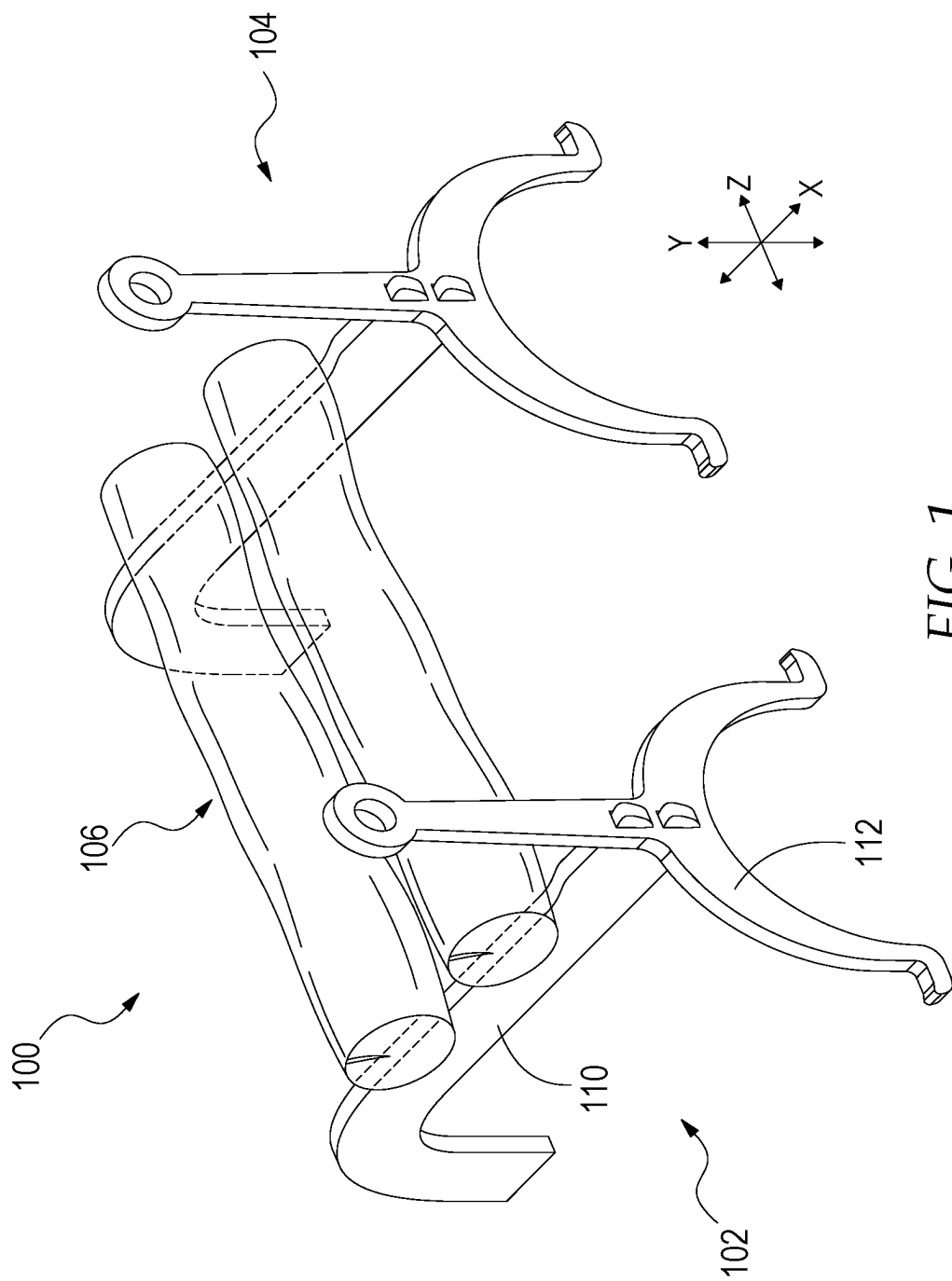
Publication Classification

(51) **Int. Cl.**
F24B 1/193 (2006.01)

(57) **ABSTRACT**

A representative andiron system includes: a first firedog exhibiting a rectangular transverse cross-section with a height dimension being at least three times longer than a width dimension, and a first front post; a front upper finger and a front lower finger, each extending from the first firedog; the first firedog, the front upper finger, and the front lower finger being formed of contiguous material and exhibiting a planar configuration, the first front post being formed of contiguous material and exhibiting a planar configuration; the first front post having a first upper opening and a first lower opening; in a first assembled configuration, the front upper finger extends through the first upper opening and the front lower finger extends through the first lower opening.





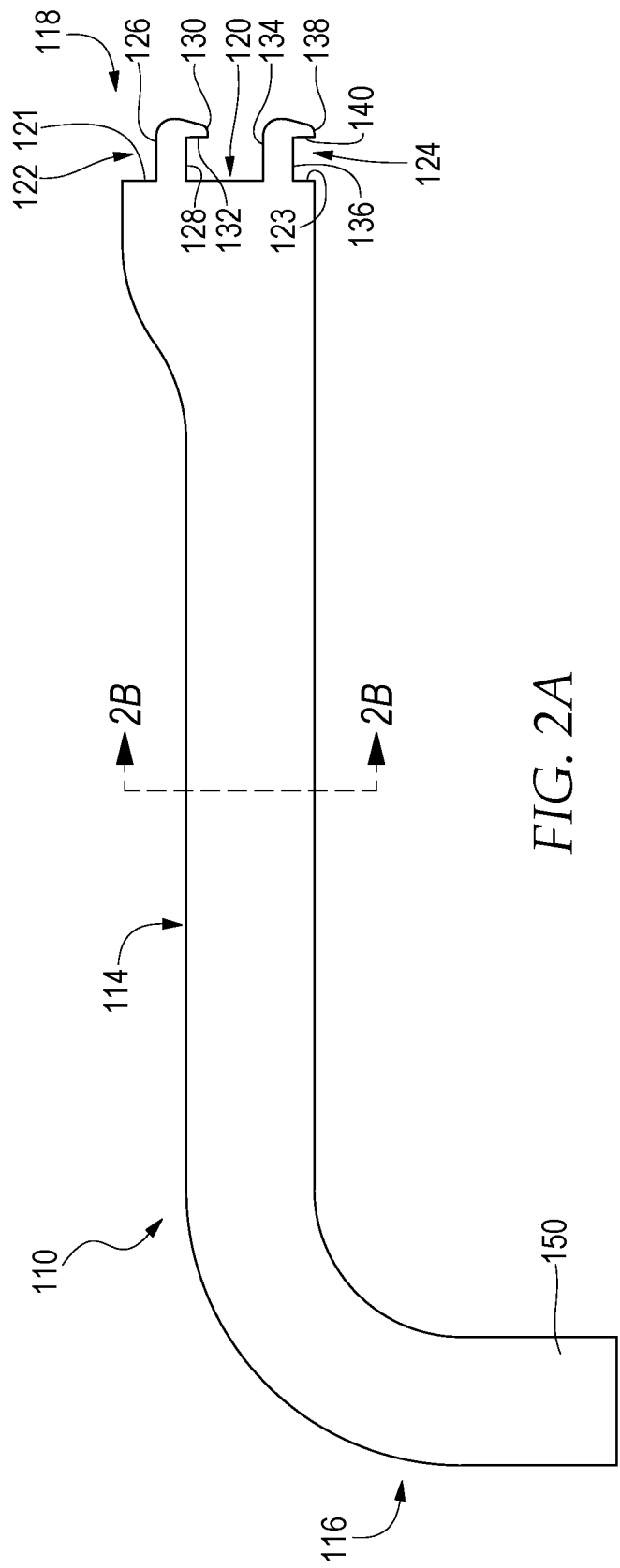


FIG. 2A

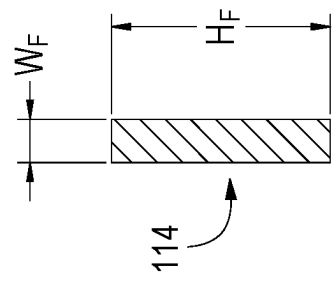
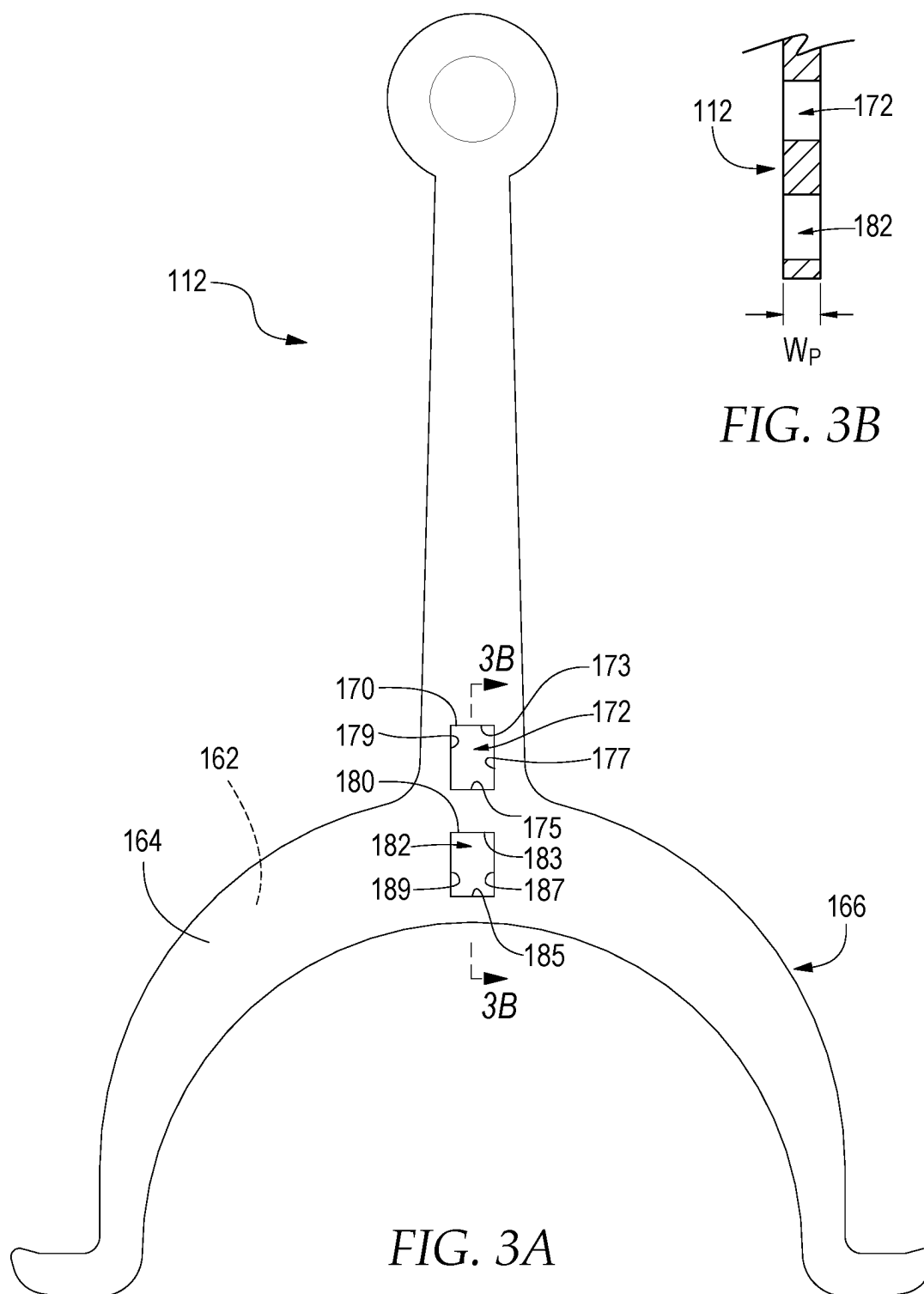


FIG. 2B



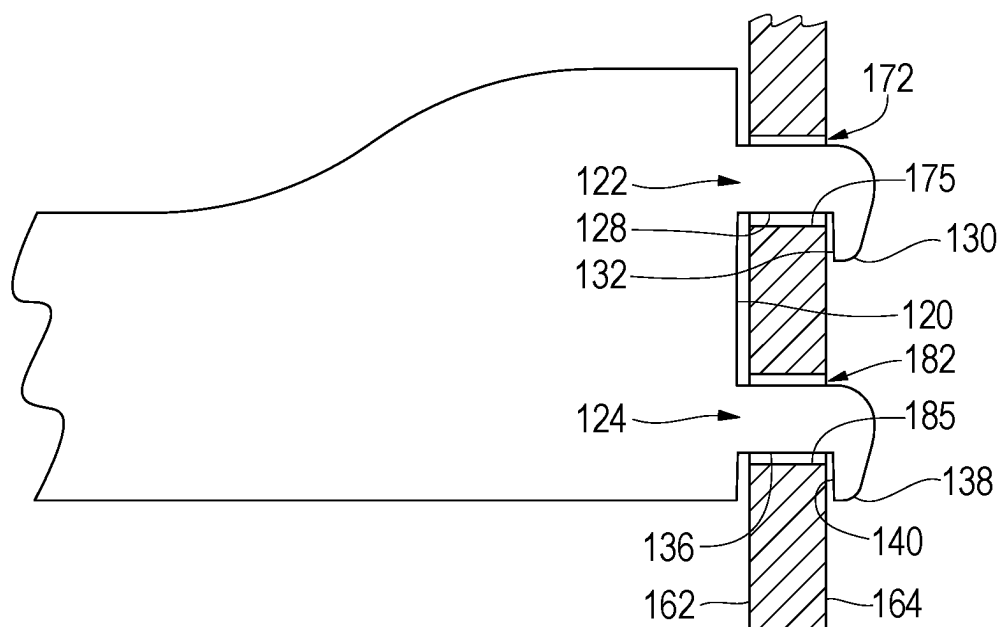


FIG. 4

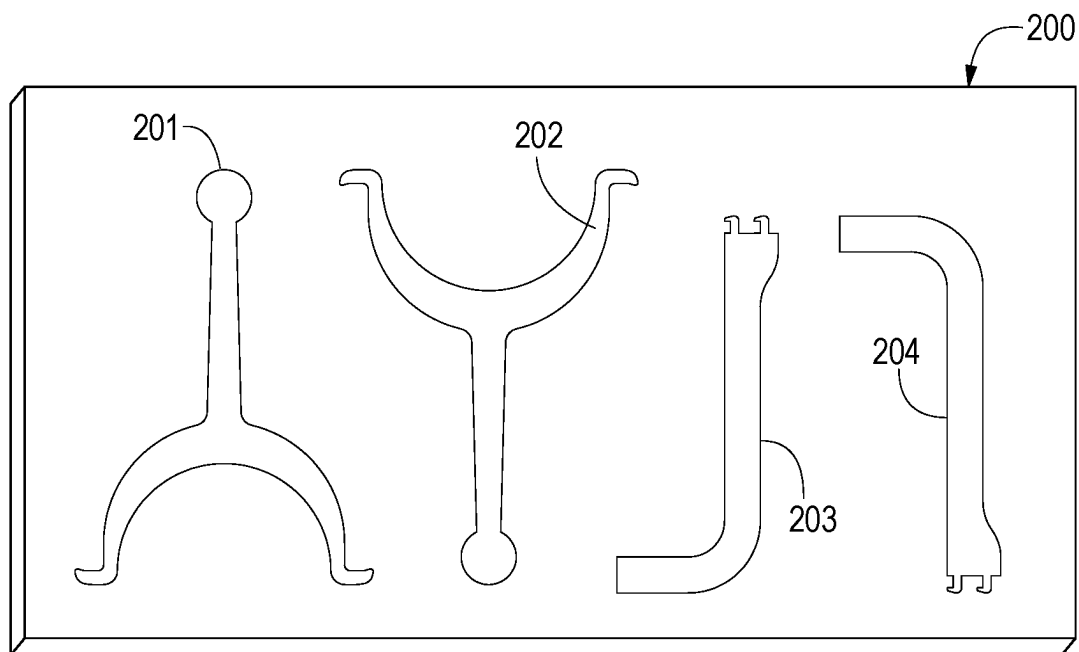
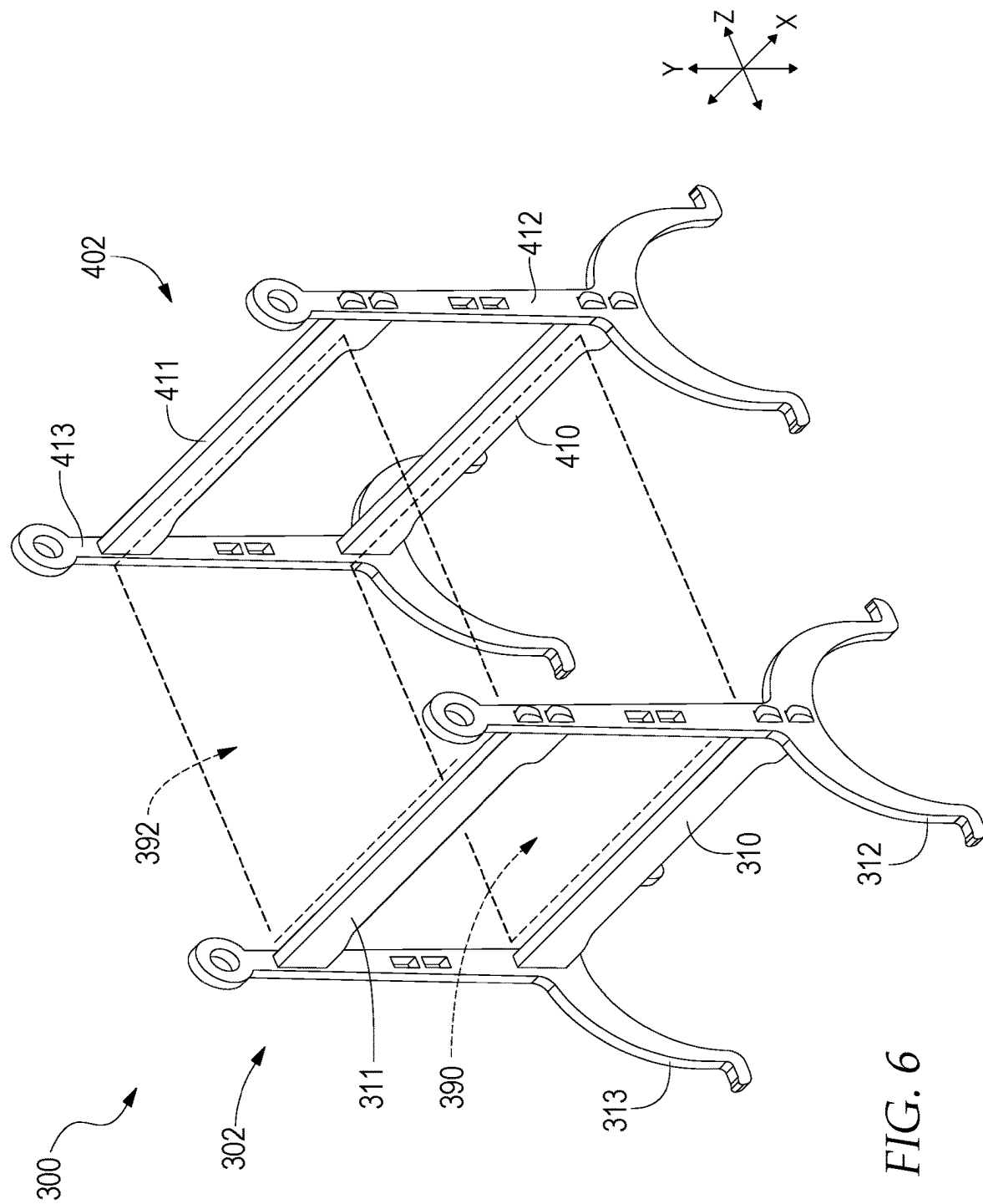


FIG. 5



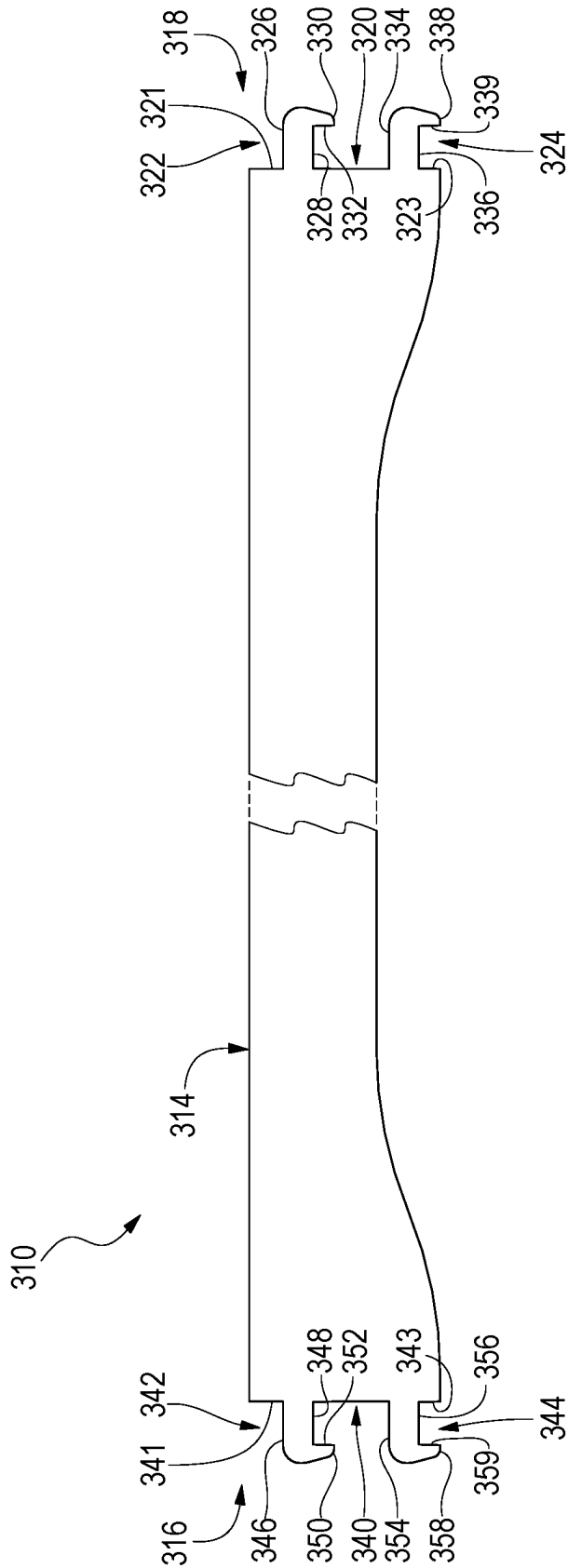


FIG. 7

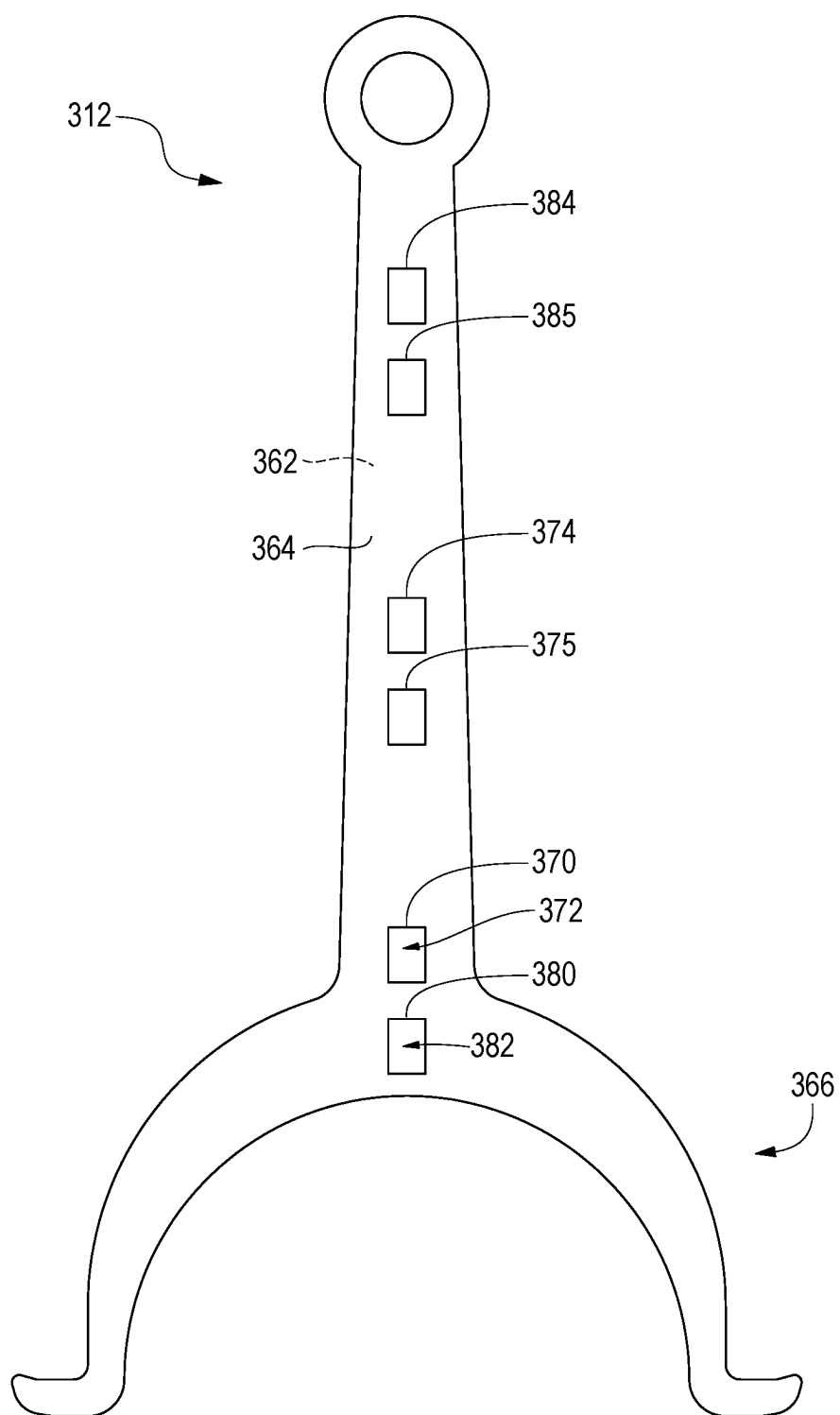
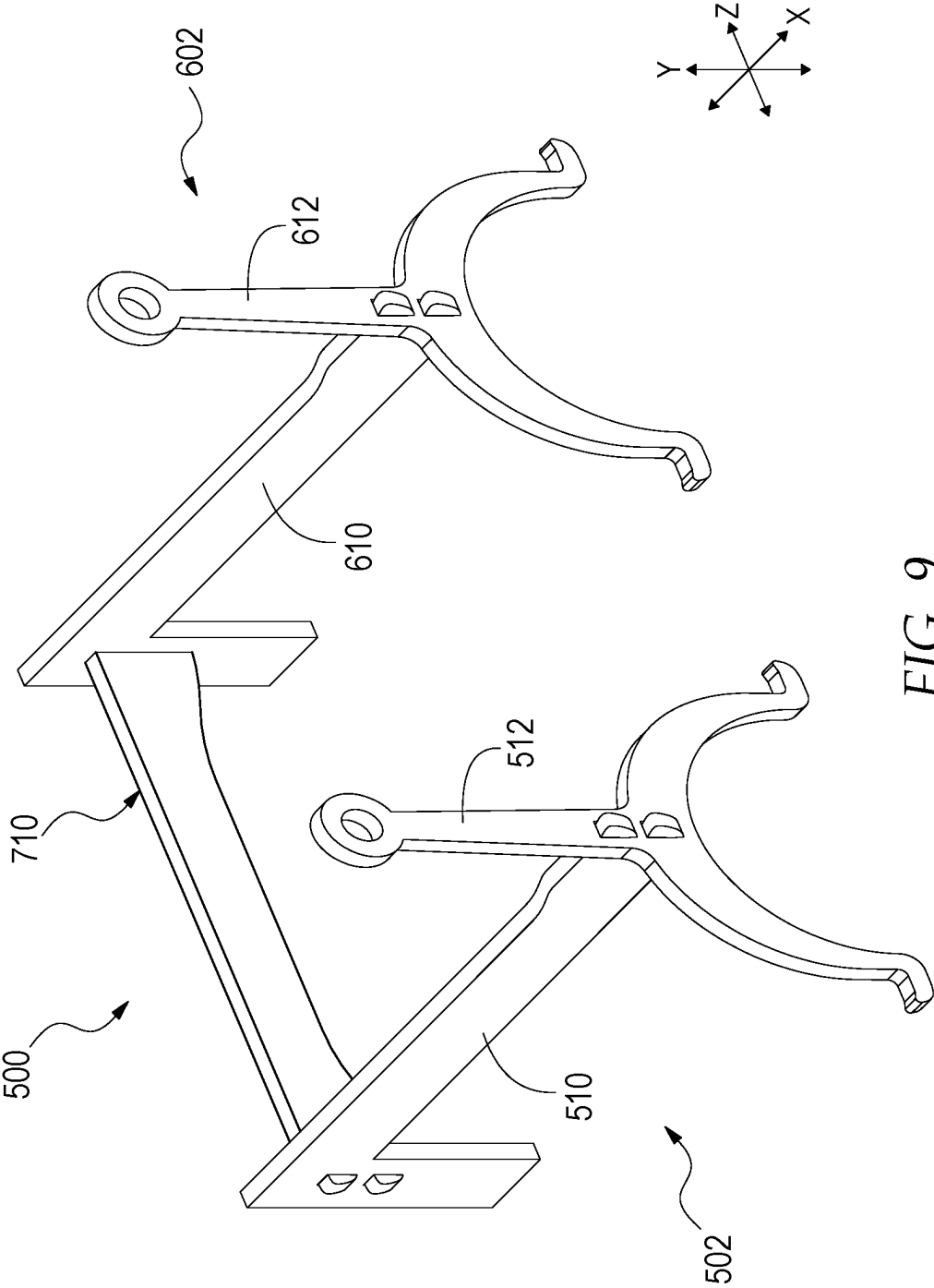


FIG. 8



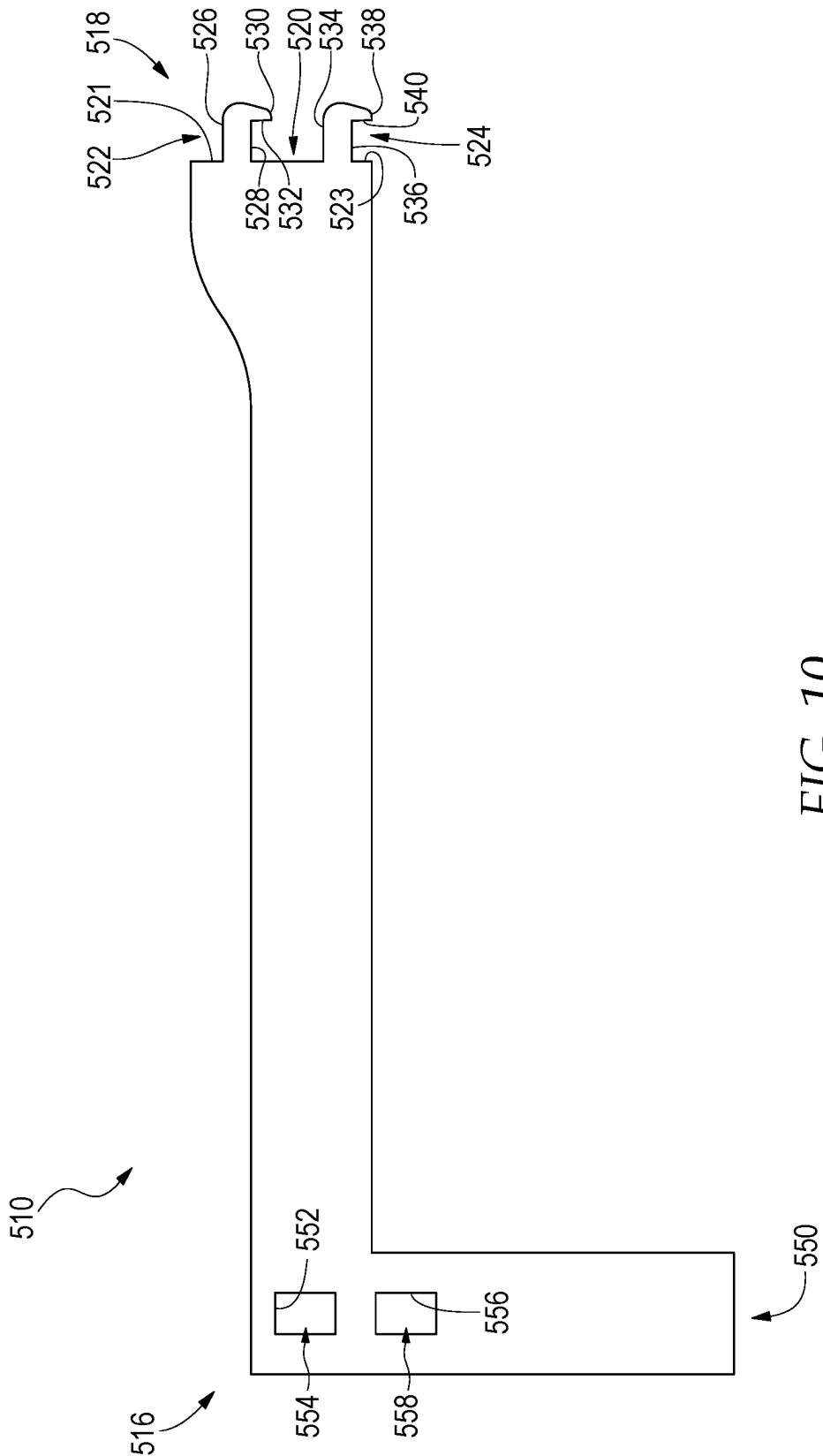
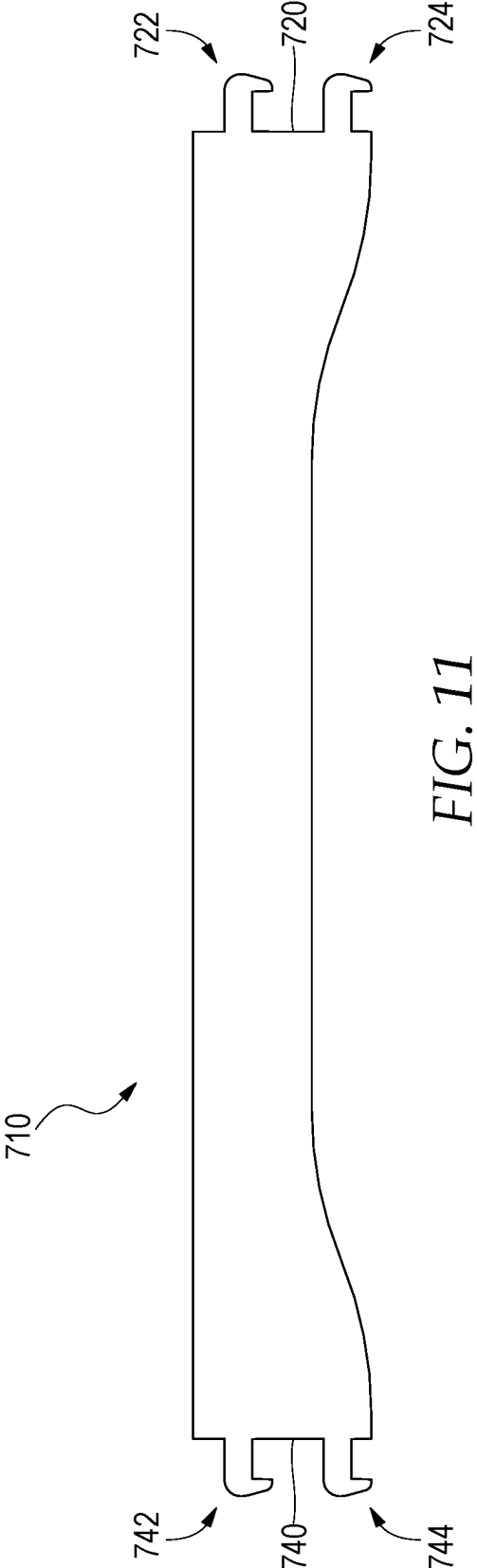


FIG. 10



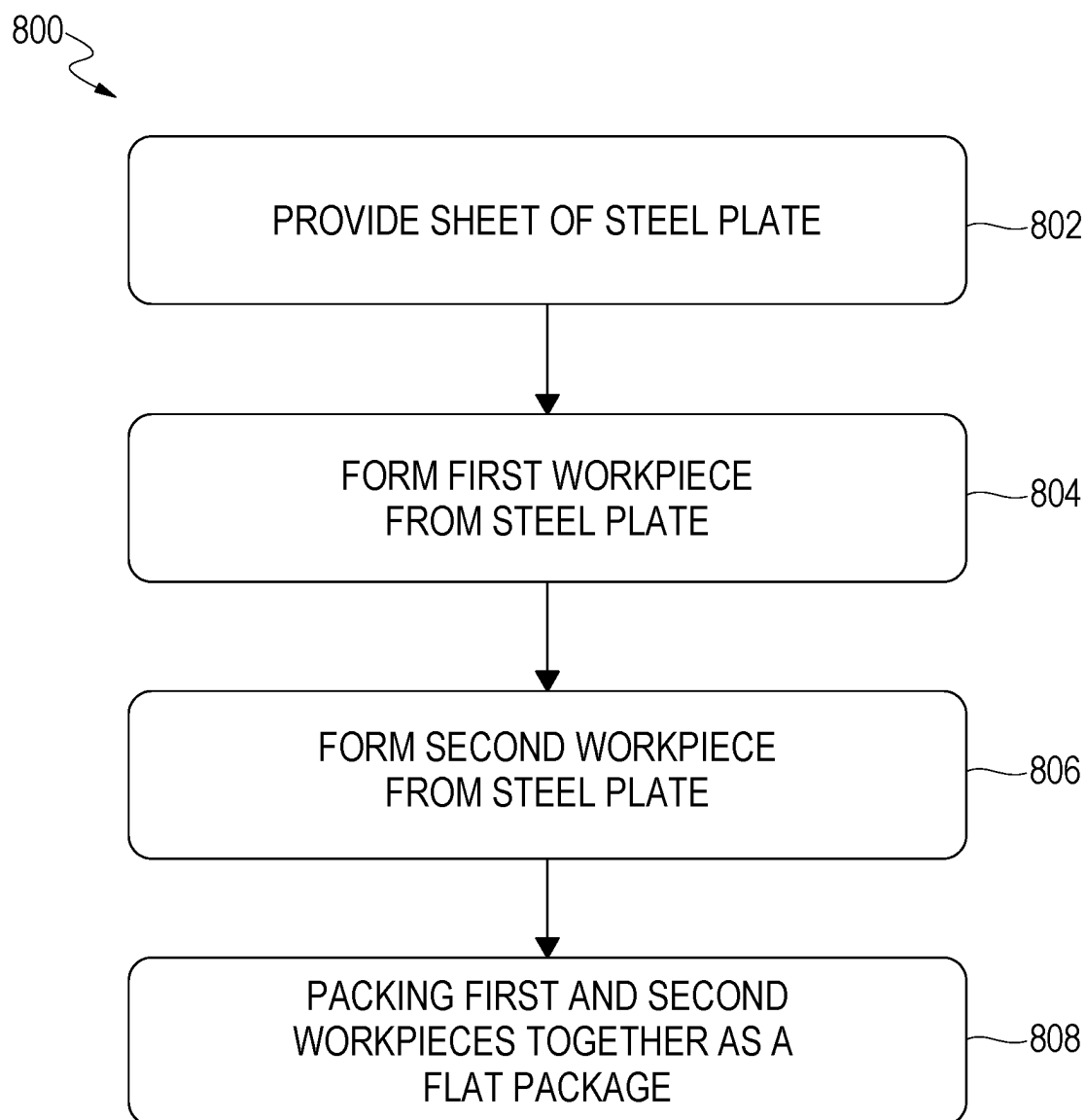


FIG. 12

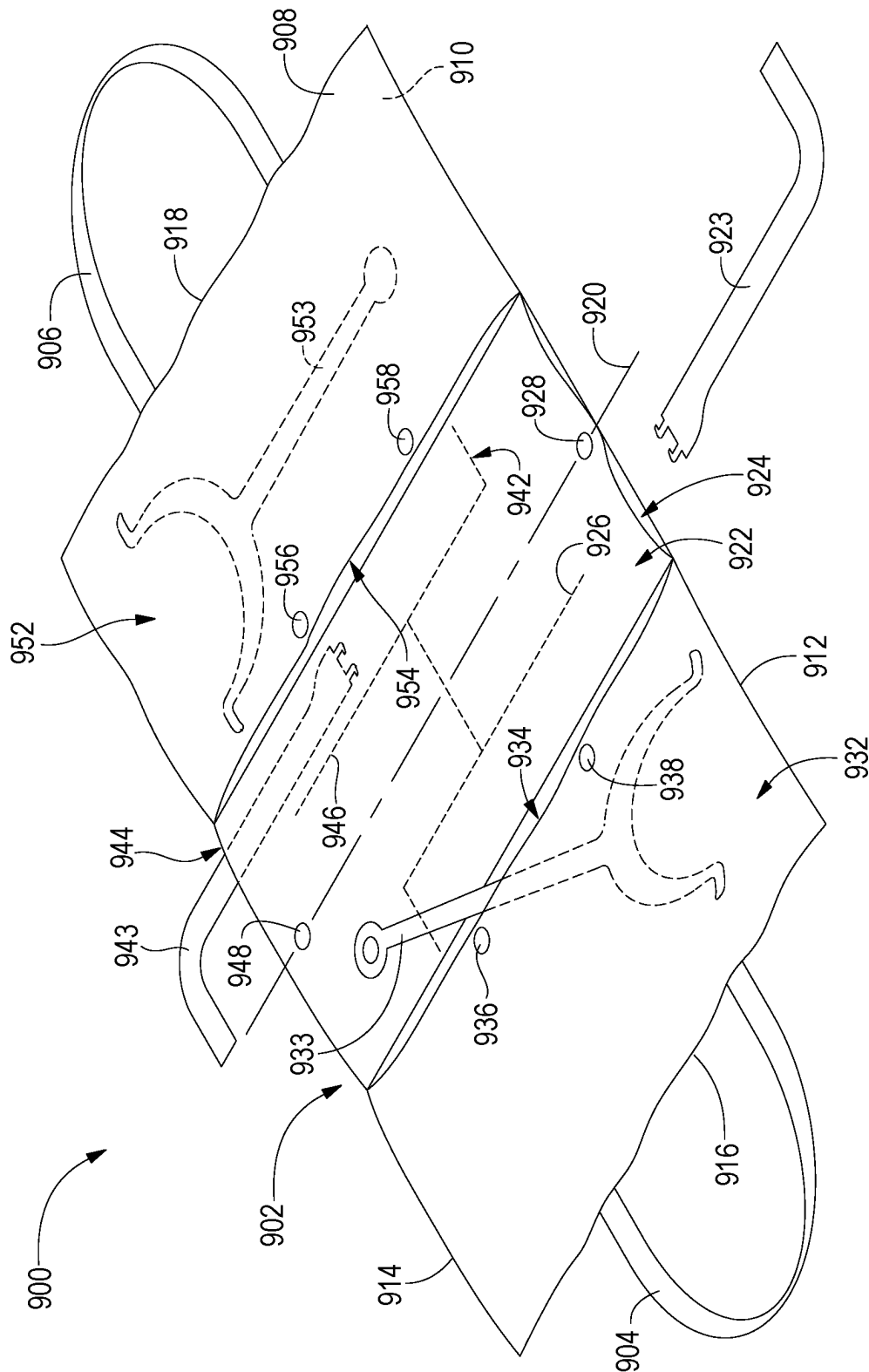


FIG. 13

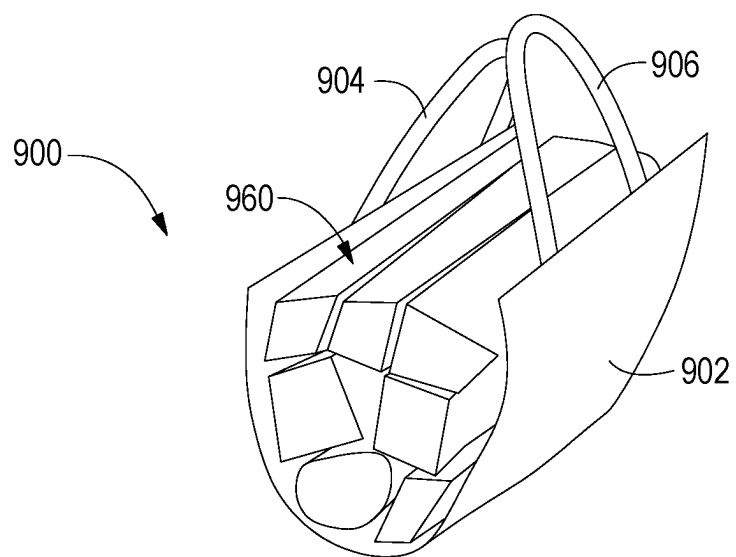


FIG. 14

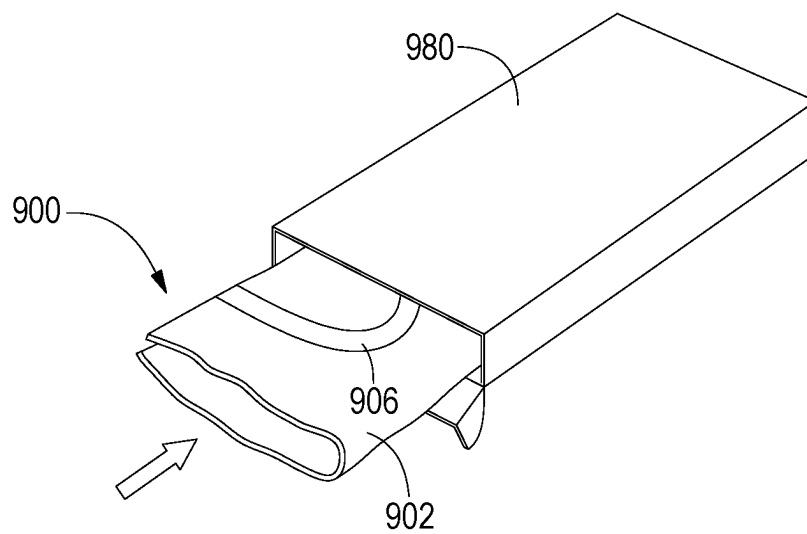


FIG. 15

INTERLOCKING ANDIRON SYSTEMS AND RELATED METHODS

BACKGROUND

Technical Field

[0001] The disclosure generally relates to andirons for use with combustible material.

Description of the Related Art

[0002] Andirons have been documented to 3 AD to elevate combustible materials (wood, for example) to enhance combustion. Andirons are the correct tool for firewood combustion. Andiron production also tends to be labor intensive owing to traditional manufacturing techniques such as forging and/or welding. Additionally, assembled andirons are bulky, resulting in logistics challenges and corresponding increases in shipping costs due to packaging constraints. As an example, conventional andirons formed of cast iron tend to be quite heavy and brittle, which may implicate significant shipping costs due to the weight as well as the amount of protective packaging (foam, for example) required.

SUMMARY

[0003] In this regard, interlocking andiron systems and related methods are provided. An example embodiment of an interlocking andiron system for use with combustible material, comprises: a first andiron assembly having a first firedog and a first front post; the first firedog having a body extending in a longitudinal direction from a proximal end to a distal end and being configured to support the combustible material in an elevated position, the first firedog exhibiting a rectangular transverse cross-section with a height dimension being at least three times longer than a width dimension, the distal end of the first firedog having a front vertical contact face; a front upper finger and a front lower finger, each extending from the front vertical contact face of the first firedog, the front upper finger having an outwardly extending front upper segment and terminating in a downwardly extending front upper tip, the front upper segment having a bottom horizontal contact surface, the front upper tip having an inwardly facing vertical contact surface, the front lower finger having an outwardly extending front lower segment and terminating in a downwardly extending front lower tip, the front lower segment having a bottom horizontal contact surface, the front lower tip having an inwardly facing vertical contact surface; the first firedog, the front upper finger, and the front lower finger being formed of contiguous material and exhibiting a planar configuration, the first front post being formed of contiguous material and exhibiting a planar configuration; the first front post having an inward face, an outward face opposing the inward face, a first upper aperture, and a first lower aperture, the first upper aperture defining a first upper opening extending through the first front post from the inward face to the outward face and being aligned to receive the front upper finger, the first lower aperture defining a first lower opening extending through the first front post from the inward face to the outward face and being aligned to receive the front lower finger, the first upper aperture having a lower horizontal contact surface, the first lower aperture having a lower horizontal contact surface; in a first assembled configuration, in which the first front post is disposed in a

vertical orientation and the body of the first firedog is disposed in a horizontal orientation, the front upper finger extends through the first upper opening with the front upper tip being disposed adjacent the outward face of the first front post, the front lower finger extends through the first lower opening with the front lower tip being disposed adjacent the outward face of the first front post, the bottom horizontal contact surface of the front upper segment engages the lower horizontal contact surface of the first upper aperture, and the bottom horizontal contact surface of the front lower segment engages the lower horizontal contact surface of the first lower aperture.

[0004] An example embodiment of a method for providing an andiron system comprises: providing a sheet of steel plate; forming, as a contiguous non-weld first workpiece, a first firedog, a front upper finger, and a front lower finger from the sheet of steel plate; and forming, as a contiguous non-weld second workpiece, a first front post.

[0005] Another example embodiment of a method for providing an andiron system comprises: providing a sheet of steel plate; forming, as a contiguous non-weld first workpiece, a first firedog, a front upper finger, and a front lower finger from the sheet of steel plate; and forming, as a contiguous non-weld second workpiece, a first front post; wherein: the first firedog has a body extending in a longitudinal direction from a proximal end to a distal end and being configured to support the combustible material in an elevated position, the first firedog exhibiting a rectangular transverse cross-section with a height dimension being at least three times longer than a width dimension, the distal end of the first firedog having a front vertical contact face; the front upper finger and the front lower finger each extend from the front vertical contact face of the first firedog, the front upper finger having an outwardly extending front upper segment and terminating in a downwardly extending front upper tip, the front upper segment having a bottom horizontal contact surface, the front upper tip having an inwardly facing vertical contact surface, the front lower finger having an outwardly extending front lower segment and terminating in a downwardly extending front lower tip, the front lower segment having a bottom horizontal contact surface, the front lower tip having an inwardly facing vertical contact surface; the first front post has an inward face, an outward face opposing the inward face, a first upper aperture, and a first lower aperture, the first upper aperture defining a first upper opening extending through the first front post from the inward face to the outward face and being aligned to receive the front upper finger, the first lower aperture defining a first lower opening extending through the first front post from the inward face to the outward face and being aligned to receive the front lower finger, the first upper aperture having a lower horizontal contact surface, the first lower aperture having a lower horizontal contact surface; in a first assembled configuration, in which the first front post is disposed in a vertical orientation and the body of the first firedog is disposed in a horizontal orientation, the front upper finger extends through the first upper opening with the front upper tip being disposed adjacent the outward face of the first front post, the front lower finger extends through the first lower opening with the front lower tip being disposed adjacent the outward face of the first front post, the bottom horizontal contact surface of the front upper segment engages the lower horizontal contact surface of the first upper aperture, and the bottom horizontal contact surface of

the front lower segment engages the lower horizontal contact surface of the first lower aperture.

[0006] Other objects, features, and/or advantages will become apparent from the following detailed description of the preferred but non-limiting embodiments. The following description is made with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Non-limiting and non-exhaustive embodiments are described with reference to the following drawings. In the drawings, like reference numerals refer to like parts throughout the various figures unless otherwise specified. These drawings are not necessarily drawn to scale. Likewise, the relative sizes of elements illustrated by the drawings may differ from the relative sizes depicted.

[0008] The disclosure can be more fully understood by the subsequent detailed description and examples with reference made to the accompanying drawings.

[0009] FIG. 1 is a schematic diagram of an exemplary embodiment of an andiron system.

[0010] FIG. 2A is a schematic diagram of the exemplary embodiment of a firedog from FIG. 1.

[0011] FIG. 2B is a cross-section of the embodiment of the firedog of FIG. 2A as viewed along section line 2B-2B.

[0012] FIG. 3A is a schematic diagram of the exemplary embodiment of a front post from FIG. 1.

[0013] FIG. 3B is a cross-section of the embodiment of the post of FIG. 3A as viewed along section line 3B-3B.

[0014] FIG. 4 is a schematic, cross-sectional cut-away showing engagement of a firedog and corresponding front post from FIG. 1.

[0015] FIG. 5 is a schematic, plan view of an exemplary embodiment of a sheet of material with an exemplary workpiece layout prior to separating the workpieces from the sheet material.

[0016] FIG. 6 is a schematic diagram of another exemplary embodiment of an andiron system.

[0017] FIG. 7 is a schematic diagram of another exemplary embodiment of a firedog.

[0018] FIG. 8 is a schematic diagram of another exemplary embodiment of a post.

[0019] FIG. 9 is a schematic diagram of another exemplary embodiment of an andiron system.

[0020] FIG. 10 is a schematic diagram of another exemplary embodiment of a firedog.

[0021] FIG. 11 is a schematic diagram of an exemplary embodiment of a cross member.

[0022] FIG. 12 is a flowchart depicting an exemplary embodiment of a method for providing an andiron system.

[0023] FIG. 13 is a schematic diagram of an exemplary embodiment of a firewood and andiron carrier.

[0024] FIG. 14 is a schematic diagram of embodiment of FIG. 13 in a carry configuration.

[0025] FIG. 15 is a schematic diagram of the embodiment of FIG. 13 in a stowed configuration being disposed within a flat pack for shipping.

DETAILED DESCRIPTION

[0026] The following describes several embodiments of interlocking andiron assemblies and related methods. It is to be understood that the invention is not limited in its application to the details of the arrangements shown since the

invention is capable of other embodiments. Also, the terminology used herein is for the purpose of description and not of limitation.

[0027] Reference throughout this specification to “one embodiment”, “an embodiment”, “one example”, and/or “an example” (or language similar thereto) means that a particular feature, structure, and/or characteristic described in connection with the embodiment or example is included in at least one embodiment. Thus, appearances of the phrases “in one embodiment”, “in an embodiment”, “one example”, and/or “an example” in various places throughout this specification are not necessarily all referring to the same embodiment or example. Furthermore, the feature(s), structure(s), and/or characteristic(s) may be combined in any suitable combination(s) and/or sub-combination(s) in one or more embodiments or examples.

[0028] In this regard, FIG. 1 depicts an exemplary embodiment of an interlocking andiron system **100** that includes two andiron assemblies (**102**, **104**) formed of steel. In FIG. 1, andiron assemblies **102**, **104** are positioned side-by-side to support combustible materials **106** (e.g., firewood) in an elevated and generally horizontal (X-Z plane) orientation above a support surface (the ground or a fireplace floor, for example). Since each andiron assembly **102**, **104** incorporates similar components/features as the other, only andiron assembly **102** will be described in detail for brevity.

[0029] Andiron assembly **102** incorporates a firedog **110** and a front post **112** that are selectively interlocked to form a freestanding structure. As shown in FIG. 2A, firedog **110** includes a body **114** that extends in a longitudinal (X axis) direction from a proximal end **116** to a distal end **118**. Firedog **110** (see, FIG. 2B) exhibits a rectangular transverse (Z axis) cross-section with a height (Y axis) dimension (H_F) being at least three times longer than a width dimension (W_F). So configured, the firedog exhibits adequate structural strength to avoid sagging under an anticipated heated load, which is a common deficiency of conventional andirons. In some embodiments, the height dimension is at least four times longer than the width dimension.

[0030] Distal end **118** includes a front vertical contact face **120**. A front upper finger **122** and a front lower finger **124** extend from front vertical contact face **120**. An upper portion **121** of front vertical contact face **120** extends upwardly from front upper finger **122**, and a lower portion **123** of front vertical contact face **120** extends downwardly from front lower finger **124**.

[0031] Front upper finger **122** includes an outwardly extending front upper segment **126** with a bottom horizontal contact surface **128**. Front upper finger **122** terminates in a downwardly extending front upper tip **130** with an inwardly facing vertical contact surface **132**. Front lower finger **124** includes an outwardly extending front lower segment **134** with a bottom horizontal contact surface **136**. Front lower finger **124** terminates in a downwardly extending front lower tip **138** with an inwardly facing vertical contact surface **140**.

[0032] A back leg **150** extends downwardly from proximal end **116**. It should be noted that various configurations of back legs (such as the depicted curved-joint configuration) may be used. In some embodiments, firedog **110**, including back leg **150**, body **114**, front upper finger **122**, and front lower finger **124** are formed of contiguous material and exhibit a planar configuration. Thus, in some embodiments, a firedog may be of uniform width along its entire length.

[0033] As shown with reference to FIGS. 3A and 3B, front post 112 includes an inward face 162 for facing combustible materials supported by firedog 110, an outward face 164 that opposes inward face 162, and a base 166. Base 166 is shown in a two-leg configuration although various other configurations may be used in other embodiments. In some embodiments, front post 112 is formed of contiguous material and exhibits a planar configuration. Thus, in some embodiments, a front post may be of uniform width (W_P) along its entire length.

[0034] A first upper aperture 170 extends through front post 112 from inward face 162 to outward face 164. First upper aperture 170 defines a first upper opening 172 that is sized, shaped, and aligned to receive front upper finger 122. First upper aperture 170 is rectangular and incorporates an upper horizontal contact surface 173, a lower horizontal contact surface 175, and side vertical contact surfaces 177, 179. A first lower aperture 180, disposed lower on first post 112 than first upper aperture 170, extends through front post 112 from inward face 162 to outward face 164. First lower aperture 180 defines a first lower opening 182 that is sized, shaped, and aligned to receive front lower finger 124. First lower aperture 180 is rectangular and incorporates an upper horizontal contact surface 183, a lower horizontal contact surface 185, and side vertical contact surfaces 187, 189.

[0035] With continued reference to FIG. 1, an assembled configuration of firedog 110 and front post 112 involves front post 112 being disposed in a vertical (Y axis) orientation and body 114 of firedog 110 being disposed in a horizontal (X-Z plane) orientation. As shown in greater detail in FIG. 4, the assembled configuration also includes front upper finger 122 extending through first upper opening 172 with front upper tip 130 being disposed adjacent outward face 164, and front lower finger 124 extending through first lower opening 182 with front lower tip 138 being disposed adjacent outward face 164. So arranged, bottom horizontal contact surface 128 of front upper segment 126 is aligned to engage lower horizontal contact surface 175 of first upper aperture 170, and bottom horizontal contact surface 136 of front lower segment 134 is aligned to engage lower horizontal contact surface 185 of first lower aperture 180. Additionally, vertical contact surfaces 132 and 140, as well as front vertical contact face 120, are positioned to restrict horizontal movement of firedog 110. It should be noted that the weight of firedog 110 itself typically is adequate to bring at least one of the bottom horizontal contact surfaces into direct contact with a corresponding lower horizontal contact surface when in the assembled configuration.

[0036] In some embodiments, the components of an andiron system may be formed from one or more sheets of material without the use of welding and/or forging. By way of example, FIG. 5 depicts an exemplary sheet of material 200 (a plate of carbon steel, for example) that may be used. Sheet of material 200 is provided with an exemplary workpiece layout that includes representative andiron system components. As shown, sheet of material 200 includes a layout for workpieces 201-204, with workpieces 201 and 203 corresponding to front post 112 and firedog 110, respectively, and workpieces 202 and 204 corresponding to the front post and the firedog of andiron assembly 104 (see FIG. 1).

[0037] After separating workpieces 201-204 from sheet of material 200, each workpiece is formed of contiguous mate-

rial and exhibits a uniform width along its length, which promotes efficient packaging and shipping. For instance, components may be stacked one atop another and, if desired, like components may be aligned. Protective packaging materials also may be reduced when compared to packaging materials typically required for multidimensional components that may incorporate protruding features that are prone to breakage.

[0038] FIG. 6 depicts another exemplary embodiment of an interlocking andiron system 300 that includes two andiron assemblies 302, 402. Andiron assemblies 302, 402 are positioned side-by-side to support combustible materials in an elevated and generally horizontal orientation. Additionally, andiron assemblies 302, 402 define supports for at least one horizontal cooking surface. Since each andiron assembly 302, 402 incorporates similar components/features as the other, only andiron assembly 302 will be described in detail aside from noting that andiron assembly 402 incorporates firedogs 410, 411, a front post 412, and a rear post 413.

[0039] Andiron assembly 302 incorporates firedogs 310, 311, a front post 312, and a rear post 313. As shown in FIG. 7, firedog 310 (which is similar to firedog 311) includes a body 314 that extends in a longitudinal direction from a proximal end 316 to a distal end 318. Firedog 310 exhibits a rectangular transverse cross-section with a height dimension (H_F) being at least three times longer than a width dimension (W_F). In some embodiments, the height dimension is at least four times longer than the width dimension.

[0040] Distal end 318 includes a front vertical contact face 320. A front upper finger 322 and a front lower finger 324 extend from front vertical contact face 320. An upper portion 321 of front vertical contact face 320 extends upwardly from front upper finger 322, and a lower portion 323 of front vertical contact face 320 extends downwardly from front lower finger 324.

[0041] Front upper finger 322 includes an outwardly extending front upper segment 326 with a bottom horizontal contact surface 328. Front upper finger 322 terminates in a downwardly extending front upper tip 330 with an inwardly facing vertical contact surface 332. Front lower finger 324 includes an outwardly extending front lower segment 334 with a bottom horizontal contact surface 336. Front lower finger 324 terminates in a downwardly extending front lower tip 338 with an inwardly facing vertical contact surface 339.

[0042] Proximal end 316 includes a back vertical contact face 340. A back upper finger 342 and a back lower finger 344 extend from back vertical contact face 340. An upper portion 341 of back vertical contact face 340 extends upwardly from back upper finger 342, and a lower portion 343 of back vertical contact face 340 extends downwardly from back lower finger 344.

[0043] Back upper finger 342 includes an outwardly extending back upper segment 346 with a bottom horizontal contact surface 348. Back upper finger 342 terminates in a downwardly extending back upper tip 350 with an inwardly facing vertical contact surface 352. Back lower finger 344 includes an outwardly extending back lower segment 354 with a bottom horizontal contact surface 356. Back lower finger 344 terminates in a downwardly extending front lower tip 358 with an inwardly facing vertical contact surface 359.

[0044] As shown in FIG. 8, front post 312 includes an inward face 362, an outward face 364, and a base 366. Base 366 is shown in a two-leg configuration although various other configurations may be used in other embodiments. In

some embodiments, front post **312** is formed of contiguous material and exhibits a planar configuration. Thus, in some embodiments, a front post may be of uniform width along its entire length.

[0045] A first upper aperture **370** extends through front post **312** from inward face **162** to outward face **164**. First upper aperture **370** defines a first upper opening **372** that is sized, shaped, and aligned to receive front upper finger **322**. A first lower aperture **380**, disposed lower on first post **312** than first upper aperture **370**, extends through front post **312** from inward face **362** to outward face **364**. First lower aperture **380** defines a first lower opening **382** that is sized, shaped, and aligned to receive front lower finger **324**.

[0046] In some embodiments, multiple apertures may be formed in a post to provide optional heights at which a firedog may be installed. In the embodiment of FIG. **8**, front post **312** incorporates two additional sets of apertures. Specifically, a second upper aperture **374** and a second lower aperture **375** are disposed higher than first upper aperture **370**, and a third upper aperture **384** and a third lower aperture **385** are disposed higher than second upper aperture **374**. Each of second upper aperture **374**, second lower aperture **375**, third upper aperture **384**, and third lower aperture **385** defines a corresponding opening that is sized, shaped, and aligned to receive a corresponding finger of a firedog. It should be noted that, in the embodiment of FIG. **6**, rear post **313** incorporates components/features similar to those exhibited by front post **312**; thus, rear post **313** will not be described in detail.

[0047] Interlocking andiron system **300** (FIG. **6**) may be provided in various assembled configurations. By way of example, front and rear posts may be positioned in vertical orientations with a corresponding firedog spanning between and interconnecting a respective one of the andiron assemblies **302**, **402**. Thus, each andiron assembly may include a single firedog disposed at a desired height based on the openings through which the corresponding fingers are inserted. As such, when the firedogs are disposed at corresponding openings of the front and rear posts of both andiron assemblies, the firedogs may be used as supports for a horizontal cooking surface such as a platform, tray, or grate spanning between the firedogs.

[0048] As another example, a different assembled configuration (as shown in FIG. **6**) includes four firedogs (firedogs **310**, **311**, **410** and **411**). In particular, firedogs **310** and **311** extend between front post **312** and rear post **313** at heights corresponding to the heights that firedogs **410** and **411** extend between front post **412** and rear post **413**. So configured, firedogs **310** and **410** may be used as supports for one horizontal cooking surface (e.g., cooking surface **390**, the location of which is shown in dashed lines and representative of one or more of various components, such as a platform, tray, and/or grate, for example). Firedogs **311** and **411** may be used as supports for another horizontal cooking surface (e.g., cooking surface **392**). Clearly, the maximum number of firedogs that may be used in forming cooking surface supports correlates with the number of openings formed through the front and rear posts.

[0049] FIG. **9** depicts another exemplary embodiment of an interlocking andiron system **500** that includes two andiron assemblies **502**, **602**. Andiron assemblies **502**, **602** are positioned side-by-side to support combustible materials in an elevated and generally horizontal orientation. Andiron system **500** also incorporates a cross member **710** that

extends between and interconnects andiron assemblies **502** and **602** to maintain spacing between the respective firedogs. Since each andiron assembly **502**, **602** incorporates similar components/features as the other, only andiron assembly **502** will be described in detail aside from noting that andiron assembly **602** incorporates firedog **610** and front post **612**.

[0050] Andiron assembly **502** incorporates a firedog **510** and a front post **512**. As shown in FIG. **10**, firedog **510** includes a body **514** that extends in a longitudinal direction from a proximal end **516** to a distal end **518**. Firedog **510** exhibits a rectangular transverse cross-section with a height dimension at least three times longer than a width dimension. In some embodiments, the height dimension is at least four times longer than the width dimension.

[0051] Distal end **518** includes a front vertical contact face **520**. A front upper finger **522** and a front lower finger **524** extend from front vertical contact face **520**. An upper portion **521** of front vertical contact face **520** extends upwardly from front upper finger **522**, and a lower portion **523** of front vertical contact face **520** extends downwardly from front lower finger **524**.

[0052] Front upper finger **522** includes an outwardly extending front upper segment **526** with a bottom horizontal contact surface **528**. Front upper finger **522** terminates in a downwardly extending front upper tip **530** with an inwardly facing vertical contact surface **532**. Front lower finger **524** includes an outwardly extending front lower segment **534** with a bottom horizontal contact surface **536**. Front lower finger **524** terminates in a downwardly extending front lower tip **538** with an inwardly facing vertical contact surface **540**.

[0053] A back leg **550** extends downwardly from proximal end **516**. Back leg **550** incorporates an upper cross-member aperture **552** that defines an upper cross-member opening **554**, and a lower cross-member aperture **556** that defines a lower cross-member opening **558**.

[0054] As shown in FIG. **11**, cross member **710** exhibits features similar to those described in relation to firedog **310** of FIG. **7**. In particular, cross member **710** includes vertical contact faces **720**, **740**, upper fingers **722**, **742**, and lower fingers **724**, **744**. Each of the fingers is configured to be received by a corresponding cross-member opening. For instance, upper cross-member opening **554** is sized, shaped, and aligned to receive upper finger **722**, and lower cross-member opening **558** is sized, shaped, and aligned to receive lower finger **724**. Additionally, cross member **710** exhibits a rectangular transverse cross-section with a height dimension at least three times longer than a width dimension. In some embodiments, the height dimension is at least four times longer than the width dimension.

[0055] FIG. **12** is a flowchart depicting an exemplary embodiment of a method for providing an andiron system. As shown in FIG. **12**, method **800** may be construed as beginning in block **802**, in which a sheet of steel plate is provided. In some embodiments, the steel plate may be carbon steel, for example. Additionally, the steel plate may be $\frac{1}{2}$ inch thick in some embodiments.

[0056] In block **804**, a first firedog with a front upper finger and a front lower finger are formed as a contiguous non-weld first workpiece from the sheet of steel plate. In some embodiments, this may be accomplished by cutting the workpiece from the steel plate, such as by waterjet cutting.

[0057] In block **806**, a first front post is formed as a contiguous non-weld second workpiece. In some embodi-

ments, the first workpiece and the second workpiece may be formed from the same sheet of steel plate.

[0058] Then, as shown in block 808, the first workpiece and the second workpiece may be packaged together for shipping as a flat package. In some embodiments, this may involve stacking the first workpiece and the second workpiece face to face.

[0059] An exemplary embodiment of a firewood and andiron carrier (hereinafter “carrier”) that may be incorporated into an andiron system is depicted in FIG. 12. Carrier 900 includes a body portion 902, a first handle 904, and a second handle 906. Each of the body portion and handles may be formed of one or more of various materials (typically selected for their strength, durability, and/or flexibility), such as canvas and leather, for example. Body portion 900 exhibits an inside surface 908 and opposing outside surface 910. Additionally, body portion 902 extends longitudinally between a front side 912 and a back side 914, and transversely between a first side 916 and a second side 918. A longitudinal axis 920 is defined along a midline of body portion 902. Each of first handle 904 and second handle 906 is elongated, with ends of first handle 904 being attached to first side 916 and ends of second handle 906 being attached to second side 918.

[0060] Body portion 902 also incorporates multiple pockets formed between layers of material forming the carrier. A first pocket 922 is configured to receive (at least partially) a first firedog 923. First pocket 922 includes a first opening 924 for providing access to an interior thereof, with first opening 924 being positioned adjacent front side 912. Stitching 926 is provided to bind overlaying layers of material forming the body portion to define the extent of first pocket 922, and one or more mechanical fasteners (928) (e.g., a snap) is provided to selectively restrict the size of first opening 924.

[0061] A second pocket 932 is configured to receive a first post 933. Second pocket 932 includes a second opening 934 for providing access to an interior thereof, with second opening 934 extending longitudinally and being disposed between first side 916 and first pocket 922. One or more mechanical fasteners (936, 938) is provided to selectively restrict the size of second opening 934, which may assist in retaining first post 933 within second pocket 932.

[0062] A third pocket 942 is configured to receive (at least partially) a second firedog 943. Third pocket 942 includes a third opening 944 for providing access to an interior thereof, with third opening 944 being positioned adjacent back side 914. Stitching 946 is provided to bind overlaying layers of material forming the body portion to define the extent of third pocket 942. One or more mechanical fasteners (948) is provided to selectively restrict the size of third opening 944.

[0063] A fourth pocket 952 is configured to receive a second post 953. Fourth pocket 952 includes a fourth opening 954 for providing access to an interior thereof, with fourth opening 954 extending longitudinally and being disposed between second side 918 and third pocket 942. One or more mechanical fasteners (956, 958) is provided to selectively restrict the size of fourth opening 954, which may assist in retaining second post 953 within fourth pocket 952.

[0064] As shown in FIG. 14, carrier 900 may be oriented in a carry configuration, in which first handle 904 and second handle 906 are disposed adjacent to each other with body portion 902 draping downwardly therebetween. So

configured, carrier 900 is suited for supporting a load upon body portion 902, such as a load of combustible materials 960 (e.g., firewood).

[0065] As mentioned previously with respect to the exemplary method of FIG. 12, workpieces may be packaged together for shipping as a flat package. In some embodiments, this may involve using a carrier (for example, carrier 900) to position and stack the workpieces. By way of example, FIG. 15 schematically depicts carrier 900 in a stowed configuration that includes firedogs and posts (not shown) being disposed within corresponding pockets of carrier 900. In the stowed configuration, carrier 900 may be disposed within a flat pack 980 for shipping. Of note, since the material layers of carrier 900 separate the firedogs and posts, use of additional packing material may be avoided, thus reducing shipping weight and/or size, which may reduce costs.

[0066] The embodiments described above are illustrative of the invention and it will be appreciated that various permutations of these embodiments may be implemented consistent with the scope and spirit of the invention as defined by the claims. Any examples provided are non-limiting examples.

I claim:

1. An interlocking andiron system for use with combustible material, comprising:

a first andiron assembly having a first firedog and a first front post;

the first firedog having a body extending in a longitudinal direction from a proximal end to a distal end and being configured to support the combustible material in an elevated position, the first firedog exhibiting a rectangular transverse cross-section with a height dimension being at least three times longer than a width dimension, the distal end of the first firedog having a front vertical contact face;

a front upper finger and a front lower finger, each extending from the front vertical contact face of the first firedog, the front upper finger having an outwardly extending front upper segment and terminating in a downwardly extending front upper tip, the front upper segment having a bottom horizontal contact surface, the front upper tip having an inwardly facing vertical contact surface, the front lower finger having an outwardly extending front lower segment and terminating in a downwardly extending front lower tip, the front lower segment having a bottom horizontal contact surface, the front lower tip having an inwardly facing vertical contact surface;

the first firedog, the front upper finger, and the front lower finger being formed of contiguous material and exhibiting a planar configuration, the first front post being formed of contiguous material and exhibiting a planar configuration;

the first front post having an inward face, an outward face opposing the inward face, a first upper aperture, and a first lower aperture, the first upper aperture defining a first upper opening extending through the first front post from the inward face to the outward face and being aligned to receive the front upper finger, the first lower aperture defining a first lower opening extending through the first front post from the inward face to the outward face and being aligned to receive the front lower finger, the first upper aperture having a lower

horizontal contact surface, the first lower aperture having a lower horizontal contact surface;

in a first assembled configuration, in which the first front post is disposed in a vertical orientation and the body of the first firedog is disposed in a horizontal orientation, the front upper finger extends through the first upper opening with the front upper tip being disposed adjacent the outward face of the first front post, the front lower finger extends through the first lower opening with the front lower tip being disposed adjacent the outward face of the first front post, the bottom horizontal contact surface of the front upper segment engages the lower horizontal contact surface of the first upper aperture, and the bottom horizontal contact surface of the front lower segment engages the lower horizontal contact surface of the first lower aperture.

2. The andiron system of claim 1, wherein an upper portion of the front vertical contact face of the first firedog extends upwardly from the front upper finger.

3. The andiron system of claim 2, wherein a lower portion of the front vertical contact face of the first firedog extends downwardly from the front lower finger.

4. The andiron system of claim 1, wherein the height dimension of the first firedog is at least four times longer than the width dimension of the rectangular transverse cross-section.

5. The andiron system of claim 1, wherein:

the proximal end of the first firedog has a back vertical contact face;

the first andiron assembly further comprises:

a back upper finger and a back lower finger, each extending from the back vertical contact face of the first firedog, the back upper finger having an outwardly extending back upper segment and terminating in a downwardly extending back upper tip, the back upper segment having a bottom horizontal contact surface, the back upper tip having an inwardly facing vertical contact surface, the back lower finger having an outwardly extending back lower segment and terminating in a downwardly extending back lower tip, the back lower segment having a bottom horizontal contact surface, the back lower tip having an inwardly facing vertical contact surface; and

a first rear post having an inward face, an outward face opposing the inward face, a first upper aperture, and a first lower aperture, the first upper aperture of the first rear post defining a first upper opening extending through the first rear post and being aligned to receive the back upper finger, the first lower aperture of the first rear post defining a first lower opening extending through the first rear post and being aligned to receive the back lower finger, the first upper aperture of the first rear post having a lower horizontal contact surface, the first lower aperture of the first rear post having a lower horizontal contact surface;

in the first assembled configuration, the back upper finger extends through the first upper opening of the first rear post with the back upper tip being disposed adjacent the outward face of the first rear post, the back lower finger extends through the first lower opening of the first rear post with the back lower tip being disposed adjacent the outward face of the first rear post, the bottom horizontal

contact surface of the back upper segment engages the lower horizontal contact surface of the first upper aperture of the first rear post, and the bottom horizontal contact surface of the back lower segment engages the lower horizontal contact surface of the first lower aperture of the first rear post.

6. The andiron system of claim 5, wherein:

the first front post has a second upper aperture and a second lower aperture, the second upper aperture and the second lower aperture being disposed higher than the first upper aperture, the second upper aperture defining a second upper opening extending through the first front post and being aligned to receive the front upper finger, the second lower aperture defining a second lower opening extending through the first front post and being aligned to receive the front lower finger;

the first rear post has a second upper aperture aligned to receive the back upper finger and a back lower aperture aligned to receive the second lower finger, the second upper aperture of the first rear post and the second lower aperture of the first rear post being disposed higher than the first upper aperture of the first rear post; and

in a second assembled configuration, the front upper finger extends through the second upper opening of the first front post, the front lower finger extends through the second lower opening of the first front post, the back upper finger extends through the second upper opening of the first rear post, and the back lower finger extends through the second lower opening of the first rear post.

7. The andiron system of claim 6, further comprising a second andiron assembly having a second front post, a second rear post, and a second firedog;

the second firedog extending between the second front post and the second rear post at a height corresponding to a height of the first firedog; and

the first firedog and the second firedog form supports for a first horizontal cooking surface.

8. The andiron system of claim 7, further comprising a third firedog and a fourth firedog;

the third firedog extending between the first front post and the first rear post at a height corresponding to the first upper opening and the first lower opening of the first front post;

the fourth firedog extending between the second front post and the second rear post at a height corresponding to a height of the third firedog; and

the third firedog and the fourth firedog form supports for a second horizontal cooking surface.

9. The andiron system of claim 1, further comprising a back leg extending downwardly from the proximal end of the first firedog.

10. The andiron system of claim 9, wherein:

the andiron system further comprises a second andiron assembly having:

a second firedog having a body extending in a longitudinal direction from a proximal end to a distal end and being configured to support the combustible material in an elevated position, the second firedog exhibiting a rectangular transverse cross-section with a height dimension being at least three times longer than a width dimension, the distal end of the second firedog having a front vertical contact face;

a front upper finger extending from the front vertical contact face of the second firedog;
 a front lower finger extending from the front vertical contact face of the second firedog;
 a back leg extending downwardly from the proximal end of the second firedog; and
 a second front post defining a first upper opening aligned to receive the front upper finger of the second firedog, and a first lower opening aligned to receive the front lower finger of the second firedog;
 in the first assembled configuration, the second front post is disposed in a vertical orientation and the body of the second firedog is disposed in a horizontal orientation, the front upper finger of the second firedog extends through the first upper opening of the second front post, and the front lower finger of the second firedog extends through the first lower opening of the second front post.

11. The andiron system of claim **10**, wherein:

the back leg of the first firedog defines a first cross-member opening;

the back leg of the second firedog defines a second cross-member opening;

the andiron system further comprises a cross member extending from a first end to a second end, the first end being configured to engage the first cross-member opening and the second end being configured to engage the second cross-member opening to maintain spacing between the first firedog and the second firedog.

12. The andiron system of claim **11**, wherein the cross member is configured as a firedog having a body extending in a longitudinal direction from a proximal end to a distal end and exhibiting a rectangular transverse cross-section with a height dimension being at least three times longer than a width dimension.

13. The andiron system of claim **1**, wherein the first firedog, the front upper finger, the front lower finger, and the first front post are cut from 1/2" steel plate.

14. The andiron system of claim **1**, wherein:

the andiron system further comprises a carrier, the carrier having a body portion, a first handle, and a second handle, the body portion extending between a front side and a back side, and between a first side and a second side, the body portion defining a first pocket and a second pocket, the first pocket configured to receive at least partially therein the first firedog, the second pocket configured to receive the first post, the first handle being elongated and having ends attached to the first side of the body portion, the second handle being elongated and having ends attached to the second side of the body portion;

wherein, in a carry configuration, the first handle and the second handle are disposed adjacent to each other with the body portion extending therebetween to form a support for carrying combustible materials.

15. The andiron system of claim **14**, wherein:

the first pocket has a first opening providing access to an interior of the first pocket, the first opening being positioned adjacent the front side of the body portion; and

the second pocket has a second opening providing access to an interior of the second pocket, the second opening being disposed between the first side of the body portion and the first pocket.

16. A method for providing an andiron system comprising:

providing a sheet of steel plate;

forming, as a contiguous non-weld first workpiece, a first firedog, a front upper finger, and a front lower finger from the sheet of steel plate; and

forming, as a contiguous non-weld second workpiece, a first front post;

wherein:

the first firedog has a body extending in a longitudinal direction from a proximal end to a distal end and being configured to support the combustible material in an elevated position, the first firedog exhibiting a rectangular transverse cross-section with a height dimension being at least three times longer than a width dimension, the distal end of the first firedog having a front vertical contact face;

the front upper finger and the front lower finger each extend from the front vertical contact face of the first firedog, the front upper finger having an outwardly extending front upper segment and terminating in a downwardly extending front upper tip, the front upper segment having a bottom horizontal contact surface, the front upper tip having an inwardly facing vertical contact surface, the front lower finger having an outwardly extending front lower segment and terminating in a downwardly extending front lower tip, the front lower segment having a bottom horizontal contact surface, the front lower tip having an inwardly facing vertical contact surface;

the first front post has an inward face, an outward face opposing the inward face, a first upper aperture, and a first lower aperture, the first upper aperture defining a first upper opening extending through the first front post from the inward face to the outward face and being aligned to receive the front upper finger, the first lower aperture defining a first lower opening extending through the first front post from the inward face to the outward face and being aligned to receive the front lower finger, the first upper aperture having a lower horizontal contact surface, the first lower aperture having a lower horizontal contact surface;

in a first assembled configuration, in which the first front post is disposed in a vertical orientation and the body of the first firedog is disposed in a horizontal orientation, the front upper finger extends through the first upper opening with the front upper tip being disposed adjacent the outward face of the first front post, the front lower finger extends through the first lower opening with the front lower tip being disposed adjacent the outward face of the first front post, the bottom horizontal contact surface of the front upper segment engages the lower horizontal contact surface of the first upper aperture, and the bottom horizontal contact surface of the front lower segment engages the lower horizontal contact surface of the first lower aperture.

17. The method of claim **16**, wherein the forming of the first workpiece comprises waterjet cutting of the steel plate.

18. The method of claim **16**, further comprising packing the first workpiece and the second workpiece together for shipping as a flat package.

19. The method of claim **18**, wherein the packing comprises stacking the first workpiece and the second workpiece face to face.

20. The method of claim **18**, wherein the packing further comprises:

providing a carrier having a body portion, a first handle, and a second handle, the body portion extending between a front side and a back side, and between a first side and a second side, the body portion defining a first pocket and a second pocket, the first pocket configured to receive at least partially therein the first firedog, the second pocket configured to receive the first post, the first handle being elongated and having ends attached to the first side of the body, the second handle being elongated and having ends attached to the second side of the body; and

inserting the first firedog at least partially within the first pocket and the first post within the second pocket.

* * * * *